Bore capability review: not ready for PPAP yet, but fixable

CNC brake caliper bore Ø 42.000 ± 0.025 mm · 3 machines · Jan–Jun 2026

For the Quality Manager · prepared by Areej Alsobh · synthetic practice dataset

The answer: not capable yet, but the machines can do it

0 of 3

machines meet Ppk ≥ 1.33 over Jan–Jun

Ppk 1.10 · 0.93 · 0.96

CNC-01 · CNC-02 · CNC-03

The machines are capable short-term

Cp 1.44–1.63 and well centred: the parts made together are consistent.

Capability is lost over time

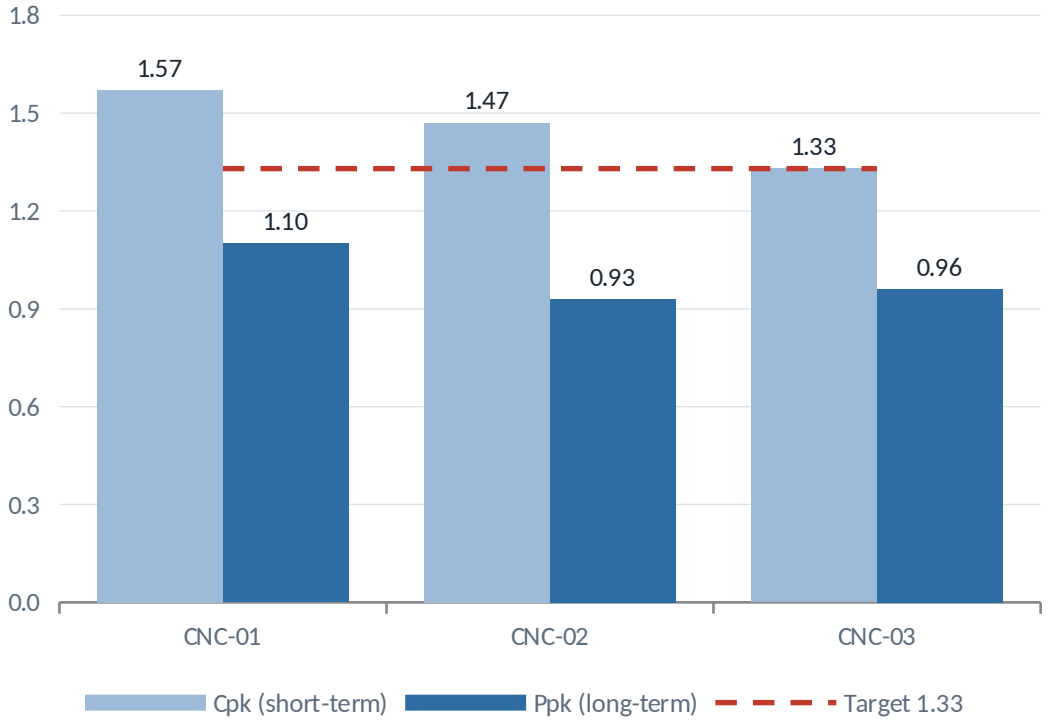
Tool wear drifts the bore 0.014 mm per tool life; 28–36% of capability is lost (Cpk → Ppk).

A fix is already proven

With tool-wear compensation, CNC-01 reached Ppk 1.55 in June.

Decision needed: approve the roll-out of tool-wear compensation to CNC-02 and CNC-03.

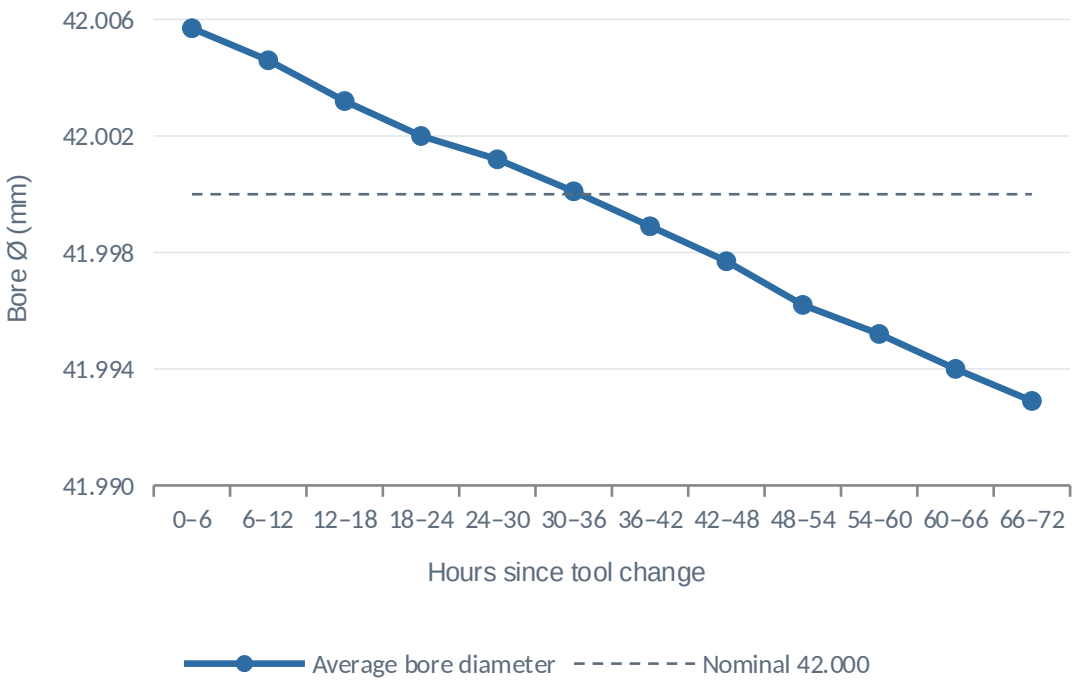
Every machine loses capability over time: Cpk passes, Ppk fails



What the gap means

- Cpk uses the spread of 5 parts made together: what the machine can do.
- Ppk uses all parts over six months: what the customer received.
- The process is not in statistical control, so Ppk is the honest figure for the customer.

Root cause 1: tool wear shrinks the bore 0.014 mm per tool life



0.014 mm

drift over a 72 h tool life: 28% of the tolerance

27–28%

of subgroups outside the $\bar{X}$ control limits (0.3% expected)

32 h

until a new tool's average falls below nominal

Root cause 2: a wrong offset ran CNC-02 high for 10 days



CNC-02 $\bar{X}$ chart, 9 Mar – 3 Apr · red dashed = control limits

- New tools on 16, 19 and 24 March started 0.013 mm high.
- The wear rate was normal: the setup was wrong, not the tool.
- The offset correction on 25 March fixed it mid-tool.
- 11 parts above the USL, all on those days.

Two more events: a slow reaction and a wider night-shift spread

CNC-01 tool breakage · 12 May, shift B

2.5 h from the first SPC signal to the tool change

- 15:00: $\bar{X}$ fell to 41.976 mm (LSL 41.975); $\bar{X}$ and R both signalled
- 17:30: tool replaced, process back to normal at once
- 34 undersize + 30 surface-finish defects that day

Action: stop and inspect at the first out-of-control signal

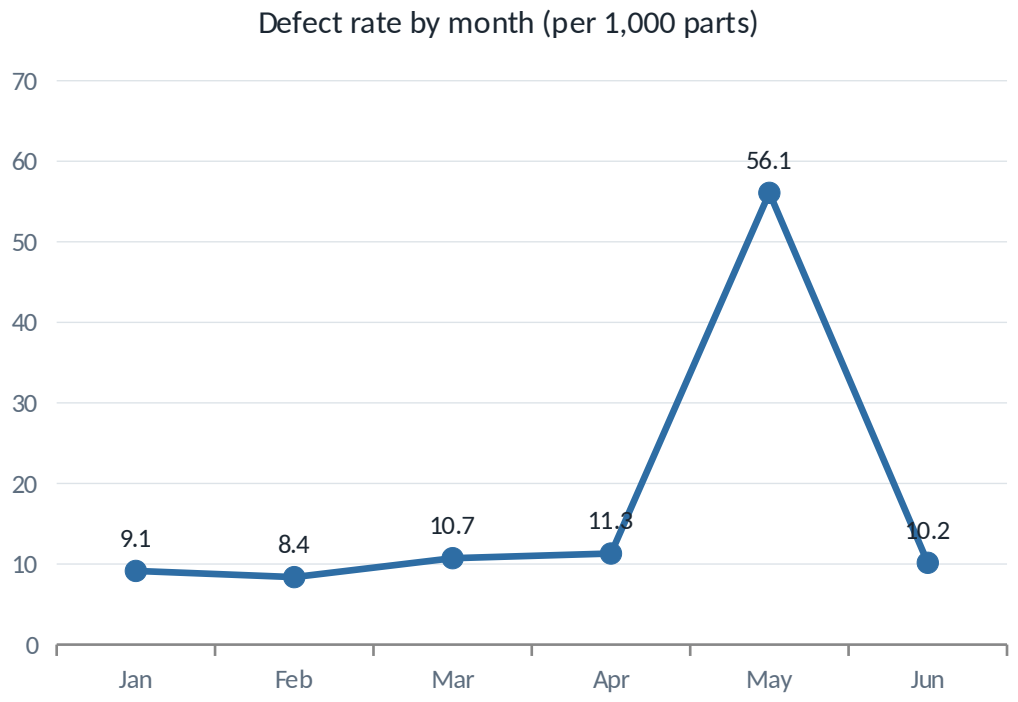
CNC-03 night shift · from 1 April

1.6× wider part-to-part spread (Levene's test, $p < 0.001$)

- Started when a new operator (6 months' experience) replaced one with 7 years
- Shifts A and B unchanged
- 20 of CNC-03's 24 out-of-spec parts

Action: onboarding and standard work, not blame

Root cause 3: one over-hard material lot caused the May spike



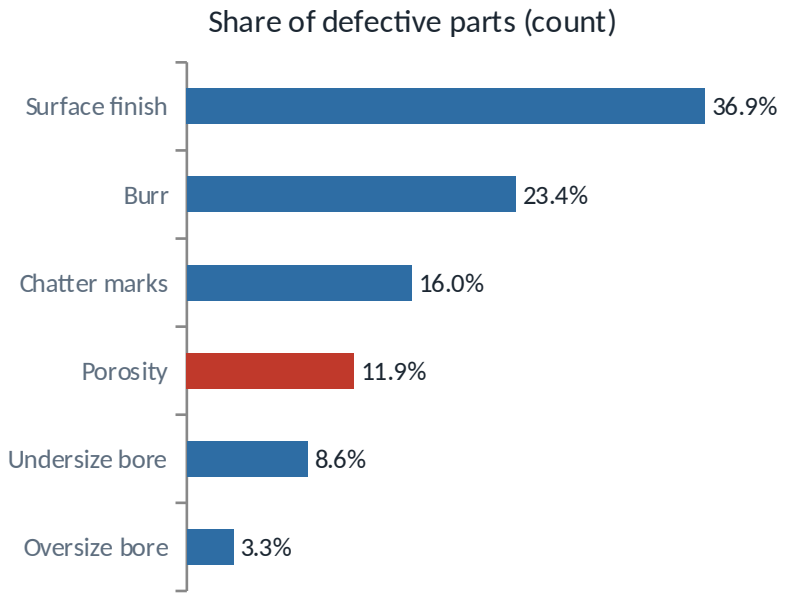
195 defects per 1,000 parts on lot L-2026-041 (other lots 7–16)

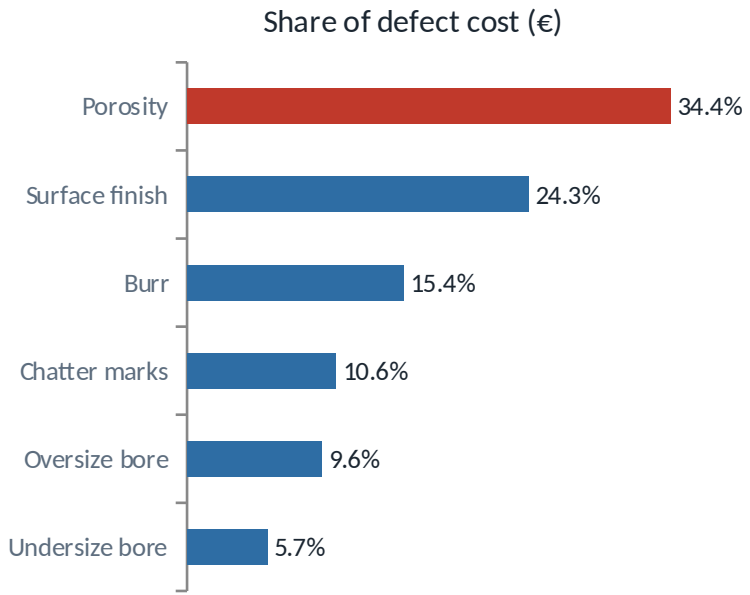
242 HB hardness of lot 041; every other lot 199–213 HB

137 of 140 rough parts (Ra > 1.6 µm) came from this one lot

Without lot 041, roughness PPU = 1.37 (passes). The same supplier (Kovar) also has chronic porosity: 3.87 per 1,000 vs 1.50 and 1.34.

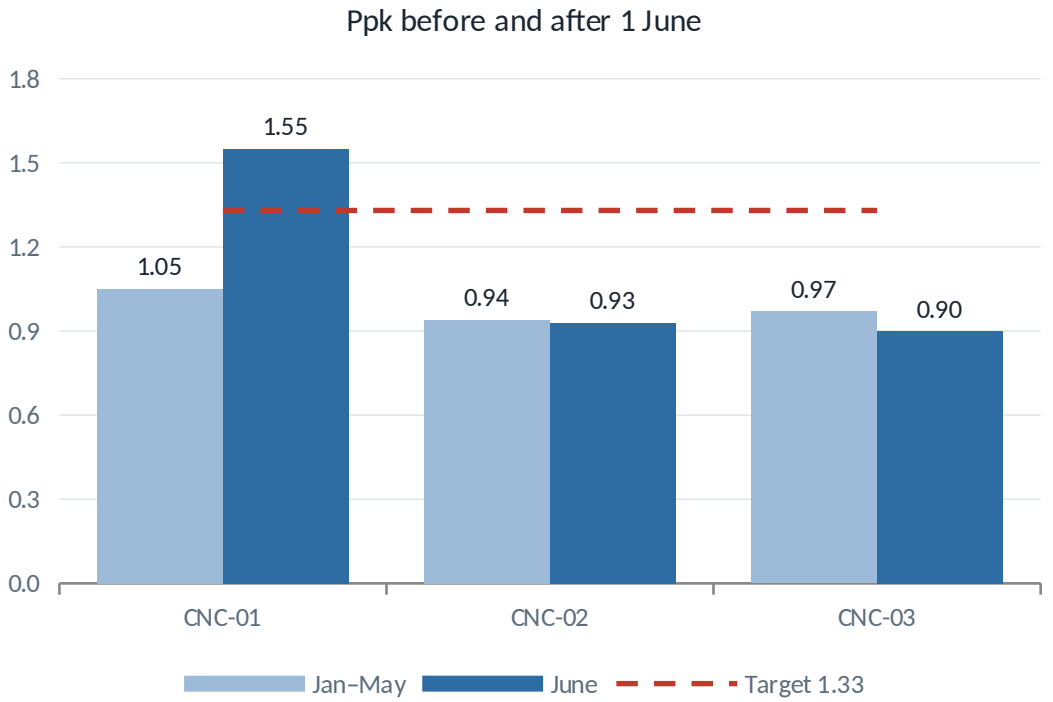
Prioritise by cost: porosity is 4th by count but 1st by cost





€28,578 total defect cost (scrap €12,580 · rework €15,998). Porosity parts are scrapped at €18.50; most other defects are reworked at €4.20.

The fix works: wear compensation lifted CNC-01 to Ppk 1.55



Why we trust it

- Overall spread −31% (Levene's test, $p < 0.001$).
- Control group: CNC-02 and CNC-03 did not improve, so it was not a good month.
- Part-to-part spread (R) unchanged: only the drift disappeared, the fingerprint of wear compensation.
- Not yet documented; one month of data.

Recommended actions, in priority order

#	Action	Addresses	Expected impact
1	Roll out tool-wear compensation to CNC-02, then CNC-03; verify each with before/after Ppk	Tool wear (all machines)	Ppk towards 1.4–1.55, as on CNC-01
2	First-part check after every tool change	Offset errors (CNC-02, March)	No more 10-day shifts after a setup error
2	Incoming hardness check with a reject limit; corrective action request to Kovar	Lot 041, chronic porosity	Roughness PPU 1.37; lower scrap cost
2	Skill matrix, work instruction and pairing before solo night shifts	CNC-03 night-shift spread	R back to ≈ 0.012 mm
3	Stop and inspect at the first SPC signal	Late reaction (CNC-01 breakage)	Contain problems hours earlier
3	Document the CNC-01 change; recalculate its control limits	Change management	Charts that detect new problems

Fallback if compensation is not possible: change tools every 48 h and set the new-tool offset ≈ 0.002 mm lower (Pp ≈ 1.41–1.44).

Path to the PPAP review

Decision today: approve the wear-compensation roll-out and the first-part check.

Weeks 1–2

Document the CNC-01 set-up; pilot on CNC-02; start first-part checks

Weeks 3–4

Roll out to CNC-03; night-shift onboarding; hardness check at goods-in

Weeks 5–8

New capability study on all machines; new control limits; PPAP submission

Appendix: data and method

Data

22,538 measurements (5 parts every 2 h per machine), production log, defect log, tool changes, material lots. Synthetic dataset built for practice.

SQL (MySQL)

Profiling and cleaning; overall quality; defect Pareto; rates by machine, shift and month; lot and supplier scorecard; tool-change windows.

Python

$\bar{X}$-R charts and Western Electric rules; Cp, Cpk, Pp, Ppk; Levene's test; correlation; before/after study; tool-wear regression.

Power BI

Star-schema model, DAX measures (rates, Cpk/Ppk, Pareto, SPC), validated against SQL and Python; interactive Overview, Pareto and SPC pages.